Latent Diffusion Research Agents: Continuous-Space Communication for Deep Research with Test-Time Diffusion

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Abstract

Deep research agents powered by large language models (LLMs) have emerged as transformative tools for automating complex, long-horizon research tasks. State-of-the-art frameworks such as the **Test-Time Diffusion Deep Researcher (TTD-DR)** achieve impressive coherence through draft-centric iterative refinement, yet their inter-agent communication remains mediated entirely by discrete token sequences - incurring information loss, representational bottlenecks and super-linear decoding costs as revision depth grows. Concurrently, **LatentMAS** has demonstrated that LLM agents can collaborate losslessly and efficiently in *continuous latent space* via KV-cache working-memory transfer, but only for short-horizon, fixed-input reasoning. We propose **Latent Diffusion Research Agents (LaDRA)**, a novel framework that *unifies* these paradigms: the multi-stage deep-research workflow of TTD-DR is augmented with latent-space inter-agent communication drawn from LatentMAS. Specifically, we replace token-mediated state transfer between planning, search-synthesis, self-evolution and report-generation agents with lossless layer-wise KV-cache propagation, preserving the full representational richness of each agent's internal reasoning across iterative diffusion cycles. We detail a concrete methodology encompassing latent-aligned denoising, latent self-evolution with fitness feedback in continuous space and a hybrid decoding strategy for external tool interaction. This work charts a principled path toward deep research agents that are simultaneously more expressive, more information-preserving and computationally more efficient than their purely text-mediated counterparts.

1 Introduction

The rapid development of LLM-based agentic systems has enabled a new class of *deep research agents* that autonomously plan, search for evidence, synthesize information and generate long-form research reports [Lu et al., 2024, Gottweis et al., 2025, Zheng et al., 2024]. Unlike single-pass generation, these agents leverage test-time compute scaling through iterative refinement loops - chain-of-thought prompting [Wei et al., 2022], self-refinement [Madaan et al., 2023], debate [Liang et al., 2023] and evolutionary optimization [Google DeepMind, 2025] - to progressively improve output quality.

A particularly compelling paradigm is the **Test-Time Diffusion Deep Researcher (TTD-DR)** [Han et al., 2025], which frames report generation as a *diffusion process*: an initial noisy draft is iteratively denoised via retrieval-augmented revision while a self-evolutionary algorithm independently optimizes each component agent (planner, question generator, answer synthesizer, report writer). TTD-DR's draft-centric design maintains global coherence and reduces information loss compared to section-wise or parallel-search architectures [Alzubi et al., 2025, ByteDance Seed, 2025].

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In parallel, the **LatentMAS** framework [Zou et al., 2025] has introduced a fundamentally different axis of improvement: replacing text-based inter-agent communication with *continuous latent-space collaboration*. In LatentMAS, each agent generates "latent thoughts" via auto-regressive last-layer hidden-state propagation and agents exchange information through layer-wise KV-cache transfer rather than decoded text. This yields lossless information preservation, $O(d_h/\log|\mathcal{V}|)$ greater expressiveness per step and 4× faster inference - but has only been validated on short-horizon, fixed-input reasoning tasks.

> **The Central Gap.** *No existing framework reconciles the representational and computational advantages of latent-space multi-agent collaboration with the long-horizon, tool-interleaved, draft-centric workflows required for deep research.* TTD-DR's inter-agent communication is mediated entirely through discrete tokens: the planner's output is decoded to text before being consumed by the search-question generator; evolved answer variants are merged as text before informing the denoiser; and the denoised draft is re-serialized at every iteration. Each such encode–decode cycle incurs **(i)** information compression through the vocabulary bottleneck, **(ii)** cumulative fidelity degradation across long revision trajectories and **(iii)** $O(|\mathcal{V}| \cdot d_h)$ decoding overhead per agent handoff. As TTD-DR's revision depth increases, these costs compound, yielding diminishing returns in quality relative to growing computational expenditure.

We propose **Latent Diffusion Research Agents (LaDRA)**, a framework that contributes:

1. **Latent-Aligned Denoising Protocol:** A mechanism that propagates the evolving draft's full hidden-state representation directly into the search-question and answer-synthesis agents via KV-cache transfer, preserving information fidelity across diffusion steps.
2. **Latent Self-Evolution:** An adaptation of TTD-DR's component-wise self-evolution that operates in continuous space - fitness evaluation and revision are performed on hidden representations, with token decoding deferred to the final cross-over step.
3. **Hybrid Decoding Gateway:** A principled boundary mechanism that selectively decodes latent states to text *only* at external tool-interaction points (e.g., web search API calls), maintaining latent communication for all internal agent handoffs.
4. **Theoretical and Empirical Analysis:** A formal characterization of information preservation and complexity gains in latent deep-research workflows, validated on standard deep-research benchmarks.

> **Aim.** To design, implement and evaluate a latent-space communication framework for draft-centric deep research agents that preserves information fidelity, improves expressiveness and reduces computational overhead across long-horizon iterative research workflows.

Specific Objectives: **O1. Formalize** the token-mediated bottleneck in TTD-DR by quantifying per-step information loss and cumulative fidelity degradation as a function of revision depth and agent count. **O2. Design** a latent working-memory transfer protocol compatible with TTD-DR's three-stage backbone (plan generation → iterative search-synthesis → report generation) and its self-evolution overlay. **O3. Develop** an input–output alignment operator (following Zou et al. [2025]) that maps last-layer hidden states back into valid input embeddings across heterogeneous agent roles without additional training. **O4. Implement** a hybrid decoding gateway that localizes text decoding to external tool boundaries, minimizing unnecessary vocabulary-space projections. **O5. Evaluate** LaDRA against TTD-DR and text-based MAS baselines on deep-research benchmarks, measuring report quality, token efficiency, inference latency and information-fidelity metrics.

2 Proposed Methodology

TTD-DR [Han et al., 2025] operates through a three-stage backbone within an outer diffusion loop. At each denoising step t, the pipeline executes: **Stage 1** (Plan Generation) → **Stage 2a** (Search Question Generation, conditioned on the current draft $\mathcal{R}_{t-1}$, query q and QA history) → **Stage 2b** (Answer Synthesis via retrieval) → **Stage 3** (Draft Revision / Report Generation). Additionally, a *self-evolution* module generates multiple variants of each stage's output, evaluates them via LLM-as-judge fitness scoring, revises based on feedback and merges the best variants. *Crucially, every inter-stage handoff currently decodes the upstream agent's output to text and re-encodes it as tokens for the downstream agent.* We propose to replace each text-mediated handoff with the latent working-memory transfer mechanism from LatentMAS [Zou et al., 2025], adapted for TTD-DR's complex workflow topology.

> **Core Mechanism.** After agent A_i completes its reasoning (Stage k), we extract its layer-wise KV-cache $\mathcal{M}_{A_i} = \{(K^{(l)}_{A_i}, V^{(l)}_{A_i}) \mid l = 1, \ldots, L\}$, which encodes both the input context and the agent's newly generated *latent thoughts* (auto-regressive last-layer hidden states). The downstream agent A_{i+1} (Stage $k{+}1$) receives $\mathcal{M}_{A_i}$ via layer-wise KV-cache concatenation, conditioning its generation on the predecessor's full internal representation without any token decoding.

Key Design Elements:

(a) Latent-Aligned Denoising. In TTD-DR's denoising loop (Algorithm 1 of Han et al. [2025]), the draft $\mathcal{R}_{t-1}$ is currently passed as text to the question generator. In LaDRA, the report-revision agent's KV-cache (containing the draft's latent representation) is directly prepended to the question generator's cache. Formally, at denoising step t: $\mathcal{M}^{(t)}_{\text{QGen}} = \textsc{Concat}(\mathcal{M}^{(t-1)}_{\text{Draft}},\ \mathcal{M}_{\text{QGen,input}})$, where $\mathcal{M}^{(t-1)}_{\text{Draft}}$ contains the full hidden-state history of the draft up to step $t{-}1$, enabling the question generator to attend over the draft's internal representation with zero information loss.

(b) Latent Self-Evolution. TTD-DR's self-evolution generates K answer variants, evaluates each via LLM-as-judge, revises and merges. In LaDRA, the initial variants are generated as latent thoughts (hidden-state sequences) and the fitness evaluator receives each variant via KV-cache transfer. Text decoding is deferred to the final *cross-over* (merge) step, where the merged output must be human-readable or must interface with the draft-revision agent: $\hat{a}^{(r)}_k = A_{\text{revise}}\left(\mathcal{M}_{a^{(r-1)}_k},\ \mathcal{M}_{\text{feedback}^{(r-1)}_k}\right), \quad r = 1, \ldots, R$, where $\hat{a}^{(r)}_k$ is the r-th revision of the k-th variant, all in latent space.

(c) Hybrid Decoding Gateway. External tools (e.g., Google Search in Stage 2b) require textual queries. LaDRA introduces a *gateway decoder* at tool-interaction boundaries that projects the question generator's latent output through the LM head: $q_{\text{search}} = \arg\max_{v \in \mathcal{V}}\ \textsc{Softmax}(h_T \cdot W_{\text{out}})$, where h_T is the final hidden state of the question-generation agent. After retrieval, the retrieved documents are re-encoded and injected into the answer-synthesis agent's KV-cache, returning to latent-space operation.

(d) Input–Output Alignment. Following Zou et al. [2025], we compute a linear alignment matrix $W_a \approx W^{-1}_{\text{out}} W_{\text{in}}$ (solved via ridge regression) to map last-layer hidden states back to valid input embeddings. This is computed once per model and reused across all latent steps and agents, adding negligible overhead.

Practical Impact: **I1. Efficiency:** By eliminating $N_{\text{agents}} \times N_{\text{steps}}$ decode–re-encode cycles, LaDRA can reduce end-to-end inference latency by an estimated 3–$4\times$ (extrapolating from LatentMAS's $4\times$ speedup on fixed-input tasks). **I2. Fidelity:** Latent transfer eliminates the vocabulary bottleneck that causes cumulative information degradation - particularly critical in TTD-DR's deep revision trajectories (10–20 denoising steps $\times$ 5–10 search iterations). **I3. Scalability:** The framework's training-free nature allows drop-in replacement of text-mediated handoffs in any existing deep-research pipeline.

> **Scientific Significance.** LaDRA provides the first principled integration of latent-space reasoning into long-horizon, tool-augmented deep research workflows. It extends LatentMAS's theoretical guarantees - lossless information transfer (Theorem 2 of Zou et al. [2025]) and $O(d_h/\log|\mathcal{V}|)$-fold expressiveness gain (Theorem 1) - from static reasoning chains to dynamic, retrieval-augmented diffusion processes. This addresses an open question at the intersection of latent reasoning [Hao et al., 2024, Zhang et al., 2025, Chen et al., 2025], multi-agent collaboration [Wu et al., 2024, Hong et al., 2023, Li et al., 2023] and deep research [Han et al., 2025, Gottweis et al., 2025].

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